M E S S O R

WHAT IS SCHEDULE OF RATES (SOR)???

- S.O.R is the list of rates of various items of work.

- To facilitate the preparation of estimates , and also to serve as guide in setting rates in connection with contract agreements.

- It is maintained in the form of printed books known as schedule of Rates (S.O.R)



M E S S O R

WHAT IS SCHEDULE OF RATES (SOR)???

It consists of group of items such as excavation items , concrete items , demolition items , plumbing items , electrification works etc.

- The S.O.R consists rates about 2500-3000 items under different subheads like Building work , Water supply , Sanitary etc.



M E S S O R

IMPORTANCE OF SOR IN ESTIMATION

Normally for arriving at cost of building two basic things are required:

1. Quantity of each and every item of work

2. Prevailing market rates of each particular item.



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The quantities can be easily worked out by an engineer based on the drawings and specifications. The rates would however, vary from place to place depending upon the rates of construction materials, labour, working environment etc.



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e.g. One institution wants to buy chairs. So first they will have to determine the specifications i.e. whether they would have of Plastic, metal etc., its dimensions, finishes. After finalizing the same, its best possible rates are estimated and the Bids by contractor will be compared with estimated rates.



M E S S O R

MES A BREIF INTRO…….

MES as a department deals with designing, construction and maintenance of buildings, road, airfields, bridges, electricity, water supply, sewerage, Sui gas, drainage, furniture and stores etc. To handle all these services, MES has different departments, like Building and Roads (B&R), Electrical and Mechanical (E&M), Furniture and Stores (F&S), which are considered to be specialists in their respective fields.



M E S S O R

MES AS AN ORGANIZATION

Military Engineer Services is organized on inter-services basis and consists of the following:-

The office of the Engineer-in-Chief.

Four MES Wings, one each for Army, Navy, PAF and Defence Production Division of Ministry of Defence, each wing is headed by a DW&CE who is of the rank of Brigadier/ Commodore/ Air Commodore /Chief Engineer.



M E S S O R

MINISTRY OF DEFENCE

MILITARY ENGINEER SERVICES

PAKISTAN

MILITARY ENGINEER SERVICES

SCHEDULE OF RATES – 2014

(Reprinted – 2015)

FOR

CONSTRUCTION AND MAINTENANCE WORKS

COMPILED UNDER THE ORDERS

OF

ENGINEER IN CHIEF

GENERAL HEADQUARTERS, RAWALPINDI



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GENERAL RULES

1. **Generally:-** Unless stated or specified elsewhere to the contrary, these General Rules shall apply to all sections, where relevant.

2. **Component Parts:-** The component parts of this Schedule (Specification, Method of Measurements, Clarification of Rates etc) shall be read together.

3. **Application:-** This schedule is intended to apply to every description of work that may be required in the construction and maintenance of buildings of all kind belonging to, hired or occupied by the Government of Pakistan (Mindef) and also include fortifications, roads, landing grounds, drainage, water supply, electrification, together with other engineering works such as sea walls, groynes, harbours, dock works, bridges and railways etc, required by various Maintenance or Special contracts.



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DESCRIPTION AND UNITS OF MEASUREMENT FOR COMMON ITEMS



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1 = SITE CLEARANCE WORKS

• This item is described in detail but the price of this item is usually indicated as lump sum (LS).

• The cost of this item is provided in the estimate by judgment, according to the description of the item and is indicated as Lump sum (L.S).



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2 = EXCAVATION FOR FOUNDATION TRENCHES

• Earthwork in excavation for foundation trenches is calculated by taking the dimensions of each trench as length X breadth X depth.

• It is measured in cubic ft, cubic yard or cubic meter, according to the prevailing practice.

• The payment for this item is generally done as Rs. per hundred cubic ft.



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MATERIAL REQUIREMENTS

1.1.2. Borrow Pits:- These are to be made only in areas and of the size and shape, approved by the Engineer-in-Charge.

1.1.3. Classification of Excavation:- For the purpose of this section, excavations shall be classified and priced under following heads:-

1.1.3.1. **Ordinary Soil** incl:-

- 1.1.3.1.1. Spoil or rubbish of every description
- 1.1.3.1.2. Rubbish tip material etc
- 1.1.3.1.3. Loose/medium stiff clay, silt, sand, gravel and loam mud
- 1.1.3.1.4. Any other formation into which a shovel can be entered with a foot pressure

1.1.3.2. **Hard Soil** incl:-

- 1.1.3.2.1. Stiff heavy clay
- 1.1.3.2.2. Mooram, Kanker
- 1.1.3.2.3. Mud concrete
- 1.1.3.2.4. Boulders that can be individually lifted by hand
- 1.1.3.2.5. Conglomerate formations, shale, lime concrete, brick work / stone masonry in lime
- 1.1.3.2.6. Metalled surface of road, bottoming
- 1.1.3.2.7. Hard core under floor
- 1.1.3.2.8. Any other formation into which a shovel cannot enter with a foot pressure and the excavation requires force-able application of a pick

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1.1.3.3. **Rock not requiring blasting** incl:-

1.1.3.3.1. CC, premix carpet, brick work and stone masonry in CM

1.1.3.3.2. Large boulders that cannot be individually lifted by hand

1.1.3.3.3. Soft rock or any other formation, which can be excavated by jumpers, wedges, hammers etc

1.1.3.4. **Rock requiring Blasting**:- This incl hard rock and similar other formations, which cannot be excavated by jumpers, wedges, hammers and light tools etc.

Note:- *Blasting shall not be performed without the prior written permission of the Engineer-in-Charge. The contractor shall take all necessary precautions for safety of persons, property etc, as required by regulations and shall implicitly observe all instructions on blasting as may be issued, without extra payment.*

1.1.3.5. **Rock requiring Blasting but Blasting prohibited**:-This incl all as in Para 1.1.3.4 above, but blasting is prohibited by the Engineer-in-Charge.

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ITEM RATES

Sl No	Description	Unit	Rate (Rs)
	Buildings		
	Foundation, Trenches, Wells etc		
1-1	**Excavation** as in **Ordinary Soil upto 1.5 M depth**, in foundation and pipe trenches upto 1.5 M wide, in shafts, wells and independent holes upto 30 sqm each and throw earth clear of edges of excavation within 10m. Timbering to be paid extra *(Foundation and Trench over 1.5 M widths will be treated as **Areas**).*	Cum	181.15
1-2	Same as item 01-1, but in **Hard Soil**.	Cum	299.66
1-3	Same as item 01-1, but in **Rock not requiring blasting.**	Cum	1166.00
1-4	Same as item 01-1, but in **Rock requiring blasting.**	Cum	1451.56
1-5	Same as item 01-1, but in **Rock requiring blasting but blasting is prohibited**	Cum	2290.35

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BRICKWORK IN FOUNDATION UP TO PLINTH

- Care must be taken, while taking dimensions from the drawings in the bill of quantities because the walls in this part of the structure are in the form of steps with changing dimensions.
- This item is calculated in cft and the unit for payment is Rs. per 100 cft.
- In the description of work, the quality of bricks and mortar ratio must be specified. For example, "Brickwork in foundation and plinth using first class bricks laid in (1:4) or (1:6) cement-sand (c/s) mortar————————“



MES SOR

SECTION – 4

BRICK MASONRY

SPECIFICATIONS

4.0. The section covers brick work and Damp Proof Course (DPC) etc.

MATERIAL REQUIREMENTS

4.1. Bricks:-

4.1.1. Sun-dried bricks shall be made of stiff clay free from stones, lime and other impurities.

4.1.2. Burnt bricks (Hand made or machine made) to be hard square and well burnt, even in size, and to be free from modules of lime. Bricks may be either sand or soap molded.

4.1.3. Fire bricks to be of best quality, country made of same size as ordinary burnt bricks, and to be of approved quality.

4.1.4. **Size of Bricks**

4.1.4.1. Sun dried bricks shall be of any size not exc (229 mm x 115 mm x 75 mm)

4.1.4.2. Hand made burnt bricks shall not be less than (220 mm x 105 mm x 67 mm)

4.1.4.3. Machine made burnt bricks shall not be less than (228 mm x 114 mm x76 mm)

4.1.5. Bricks shall be well burnt, of uniform color, regular in shape and size with square corners & parallel faces, and should emit a clear ringing sound when struck. Table 4-1 shows max limit of water absorption test and compressive strength of bricks.

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ITEM RATES

Sl No.	Description	Unit	Rate (Rs)
	Brickwork		
4-1	Sun dried brick work in mud mortar.	Cum	3958.06
4-2	Burnt brick work, straight or to curve with inner radius of 6m and over, laid and jointed in mud mortar.	Cum	8113.03
4-3	Burnt brick work, straight or to curve with inner radius of less than 6m etc, laid and jointed in mud mortar.	Cum	8548.71



M E S S O R

DAMP PROOF COURSE (D.P.C.)

 Horizontal D.P.C. shall extend the full width of the super structure walls, however, it shall not be provided across veranda openings. It is also provided in roof and floors. Vertical D.P.C. is provided in external walls, especially, in the walls of basements.

 The quantity of D.P.C. is estimated in square ft.(on area basis) and standard unit for payment is Rs. per 100 sft.



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mortar, incl preparing surface for re-pointing

Damp Proof Course (DPC)

4-98	13mm thick DPC, of CM 1:3 mixed with **Pudlo** (or other similar approved material) 3% by weight.	Sqm	350.37
4-99	Same as item 04-98, but 20mm thick.	Sqm	525.78
4-100	DPC of **Hessien Cloth # 2,** impregnated with bitumen as specified (hot application).	Sqm	280.69
4-101	DPC of double layer of **Polythene sheet**, as specified.	Sqm	495.40
4-102	50mm thick P.C.C 1:2:4, using 3mm aggregate and finished smooth with steel float, covered with two coats, each of hot bitumen 0.75 kg per Sqm and blinded with 0.012 cum of sand per Sqm.	Sqm	658.83

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M E S S O R

REINFORCEMENT STEEL / GENERAL STEEL WORK

- Steel is provided separately from R.C.C. per ton, per Kg, or per cwt (standard weight also called Quintal or century weight equal to 112 Ibs = 50Kg).

- Quantity of steel can either be worked out by rules of thumb practice or by intensive calculations taking the length and diameter of steel bars from the working drawings showing reinforcement details and bar-bending schedules. In taking length of bars, due margin of hooks, bends and overlapping is given



M E S S O R

ITEM RATES

Sl No	Description	Unit	Rate (Rs)
	Structural Steel Works		
9-1	Supply and fix, **joist, channel, angle or tee sections**, fixed independently without connecting plates etc, cut to ordinary length.	Kg	121.64
9-2	Same as item 09-1, but erection only	Kg	7.82
9-3	Add to item 09-1 or 09-2,if fixed in repairs	Kg	4.69
9-4	Supply and fix **purlins, hips and the like**, main and cross beams or grids with angle cleats or other connecting plates, cut to dead length where required.	Kg	127.66

SERVICES



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PLASTERING

- The type of plaster, proportioning of materials and minimum thickness of plaster have to be specified.

- The quantity is calculated for total wall surface without deduction for openings such as doors windows, ventilators, etc. However, if the wall is being plastered on both the faces, the deductions for opening areas are made from one side only.

- Standard unit for payment is Rs. per 100 sft.

- Height is also specified for plastering because, for greater heights, labor cost increases. The rate varies according to the number of the storey



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ITEM RATES

Sl No.	Description	Unit	Rate (Rs)
	Mud Plaster		
13-1	19mm thick **Mud Plaster** incl 3mm thick leeping	Sqm	212.23
13-2	Same as item 13-1 but 25 mm thick.	Sqm	264.78
	Cement Plaster		
13-3	13mm thick **Cement Plaster** 1:3 finished as specified.	Sqm	235.42
13-4	Add to item 13-3, for **Pudlo** or other similar approved material mixed with cement @ 3% by weight.	Sqm	33.20
13-5	13mm thick **Cement Plaster** 1:4 finished as specified.	Sqm	219.01

SERVICES



M E S S O R

WOODWORK/CARPENTRY

- The type of material used and the quantity of finish required should be clearly indicated in the description of the item.

- The rate for any type of woodwork includes cutting of timber to required sizes, joinery work, fittings and fastenings, three coats of oil paints or varnish, bolts, locks, handles, etc.

- The measurements are taken for the overall area of doors, windows, etc. If volume of timber required for these items is to be finding out, the computed area is multiplied with the nominal thickness and an allowance of 25% is made for wastage of timber.



MESSOR

Wood Work

6-4	Supply and fix **timber in scantling**, Ist Class softwood. Complete.	Cum	140681.98
6-5	Same as item 06-4, but hardwood (Shisham).	Cum	44427.98
6-6	Same as item 06-4, but Burma Teak.	Cum	348578.62
6-7	Supply and fix timber in scantling framed Ist Class softwood Complete.	Cum	153265.44
6-8	Same as item 06-7, but hardwood (Shisham).	Cum	55650.73
6-9	Same as item 06-7, but Burma teak.	Cum	387009.26
6-10	Supply and fix, timber in scantling wrought and framed, Ist Class softwood complete.	Cum	159373.04
6-11	Same as item 06-10, but hardwood (Shisham).	Cum	62369.20
6-12	Same as item 06-10, but Burma Teak.	Cum	414153.03

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M E S S O R

SOME COMMON LABOUR RATES







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SECTION 31

LABOUR RATES

These Labour Rates have been used in compiling the item rates in this Schedule and shall be used in accordance with the General Rules, where applicable.

S/No.	Description	Rate per day (Rs)
31-1	Aluminium Fixer	800.00
31-2	Black Smith	800.00
31-3	Brick-Layer (Mason)	800.00
31-4	Cabinet maker	800.00
31-5	Cable jointer	800.00
31-6	Carpenter	800.00
31-7	Carpenter boy	450.00
31-8	Chowkidar	400.00
31-9	Electrician	800.00
31-10	Fitter Mechanic	900.00
31-11	Gas Fitter	800.00

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31-12	Glazier	700.00
31-13	Hammer-man	500.00
31-14	Labour	500.00
31-15	Labour Electric	550.00
31-16	Line Man	800.00
31-17	Mali	400.00
31-18	Painter	800.00
31-19	Pipe Fitter	800.00
31-20	Plumber	800.00



M E S S O R

S/No.	Description	Rate per day (Rs)
31-21	Polisher / Grinder	800.00
31-22	Roof Tiler	800.00
31-23	Sanitary Worker / Khakrob	500.00
31-24	Sawyer	650.00
31-25	Sign-Writer	750.00
31-26	Stone Breaker	500.00
31-27	Stone Mason	800.00
31-28	Supervisor	700.00
31-29	**Tile Fixer** (for floor and wall tiles work, any type, glazed / unglazed, ceramic, porcelain, marble, granite, vinyl etc).	1000.00
31-30	Tin smith	900.00
31-31	Up-holster	800.00
31-32	Welder	800.00
31-33	Well Sinker - In charge	750.00
31-34	Wire man	800.00
31-35	X-Ray Welder	1000.00

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THANK YOU.....



DO YOU HAVE ANY QUESTIONS ?

REFERENCES :-
MES SCHEDULE OF RATES 2014-15